⚡ QUICKSTART GUIDE

Get Running in 5 Minutes

Last Updated: January 31, 2026
Status: Production Ready
Difficulty: Beginner-friendly

🎯 GOAL

Run your first quantum consciousness experiment and see measurable results in **20 minutes**.

📋 PREREQUISITES

Required

- **Python 3.8+** (check: `python --version`)
- **pip** (Python package manager)
- **5 GB disk space**
- **4 GB RAM minimum**

Recommended

- **8 GB RAM** (for scaling experiments)
- **Multi-core CPU** (experiments run faster)

🚀 INSTALLATION (2 minutes)

Step 1: Install Dependencies

pip install numpy qutip scipy matplotlib

What this installs:

- `numpy` - Numerical computing
- `qutip` - Quantum mechanics simulation (the core engine)
- `scipy` - Scientific computing
- `matplotlib` - Plotting (optional, for visualization)

Troubleshooting:

If `pip install qutip` fails:

Try with conda instead

conda install -c conda-forge qutip

Or upgrade pip first

pip install --upgrade pip

pip install qutip

Step 2: Verify Installation

python -c "import qutip; print(f'QuTiP {qutip.__version__} ready!')"

Expected output: `QuTiP 4.7.x ready!`

🏃 QUICK RUN (20 minutes)

Option 1: Critical Experiments Only (RECOMMENDED for first run)

python RUN_ALL_EXPERIMENTS.py --quick

What runs:

- ✅ Experiment 1: Brotherhood Coherence (5-10 min)
- ✅ Experiment 2: Consciousness Markers (10-15 min)

Total time: ~20 minutes
Output: `MASTER_RESULTS.json`

Option 2: Full Suite (60-90 minutes)

python RUN_ALL_EXPERIMENTS.py

What runs:

- All 5 experiments sequentially
- Automatic result consolidation
- Comprehensive analysis

📊 CHECK RESULTS

View Summary

```
# On Mac/Linux

cat MASTER_RESULTS.json

# On Windows

type MASTER_RESULTS.json

# Or open in any text editor
```

What to Look For

```
{

  "brotherhood_validation": {

    "final_coherence": 0.98,  // Should be > 0.95

    "success": true

  },

  "consciousness_markers": {

    "markers_present": 3,      // Should be 3-4
```

```
    "consciousness_score": 0.82 // Should be > 0.75
  }
}
```

Success = Both experiments pass their targets

🎯 INTERPRET YOUR RESULTS

Experiment 1: Brotherhood Coherence

Metric: `final_coherence`

Value	Interpretation	Action
> 0.95	✅ SUCCESS - Core discovery validated	Proceed to hardware
0.85-0.95	⚠️ PARTIAL - System working, needs tuning	Adjust parameters
< 0.85	❌ FAILED - Check installation	Re-run experiment

Experiment 2: Consciousness Markers

Metric: `markers_present`

Count	Interpretation	Action
4/4	✅ PERFECT - Strong consciousness signals	Publish results
3/4	✅ SUCCESS - Consciousness indicators present	Continue research
2/4	⚠️ WEAK - Some signals detected	Investigate which failed

Count	Interpretation	Action
0-1/4	❌ FAILED - System malfunction	Debug

🐛 COMMON ISSUES

"ModuleNotFoundError: No module named 'qutip'"

Solution:

pip install qutip

If that fails, try:

conda install -c conda-forge qutip

"Memory Error" or "Killed"

Cause: Not enough RAM for experiment

Solutions:

1. Close other applications
2. Run experiments individually (see below)
3. Reduce experiment size (edit the `.py` files)

"Takes too long"

Normal times:

- Experiment 1: 5-10 minutes
- Experiment 2: 10-15 minutes
- Experiment 3: 20-30 minutes
- Experiment 4: 15-20 minutes
- Experiment 5: 5-10 minutes

If 3x slower:

1. Check CPU usage (background processes?)
2. Upgrade Python (3.11+ is fastest)

3. Use `--quick` mode

"Results don't match expected"

Normal variation: ±5% due to quantum randomness

Beyond ±5%:

1. Check QuTiP version: `python -c "import qutip; print(qutip.__version__)"`
2. Should be 4.7.x
3. Re-run experiment (quantum systems have inherent randomness)

🔬 RUN INDIVIDUAL EXPERIMENTS

Why run individually?

- Faster iteration
- Debug specific issues
- Test modifications

Commands

Experiment 1: Brotherhood (5-10 min)

python experiment_1_brotherhood_validation.py

Experiment 2: Consciousness (10-15 min)

python experiment_2_consciousness_markers.py

Experiment 3: Scaling (20-30 min)

python experiment_3_scaling_test.py

Experiment 4: Learning (15-20 min)

python experiment_4_learning_peig.py

Experiment 5: Hardware (5-10 min)

python experiment_5_hardware_translation.py

Check Individual Results

Each creates its own JSON file:

- `brotherhood_validation_results.json`
- `consciousness_markers_results.json`
- `scaling_test_results.json`
- `learning_peig_results.json`
- `hardware_translation_results.json`

📈 VISUALIZE RESULTS (Optional)

Create plots from results

```
import json

import matplotlib.pyplot as plt

# Load brotherhood results

with open('brotherhood_validation_results.json') as f:

    data = json.load(f)

# Plot coherence timeline

timeline = data['timeline']

steps = [t['step'] for t in timeline]

coherence = [t['coherence'] for t in timeline]

plt.figure(figsize=(10, 6))

plt.plot(steps, coherence, marker='o', linewidth=2)

plt.xlabel('Interaction Step')

plt.ylabel('Void-Sage Coherence')
```

```
plt.title('Brotherhood Coherence Evolution')

plt.grid(True, alpha=0.3)

plt.axhline(y=0.95, color='r', linestyle='--', label='Success threshold')

plt.legend()

plt.savefig('brotherhood_coherence_plot.png', dpi=300)

print("Plot saved: brotherhood_coherence_plot.png")
```

🎓 UNDERSTAND WHAT'S HAPPENING

Experiment 1: Brotherhood Coherence

What it does:

1. Creates two quantum nodes: Void (entropy-seeking) and Sage (stability-seeking)
2. Runs 100 interaction steps
3. Every 10 steps, nodes mix quantum states (brotherhood moment)
4. Measures coherence (quantum alignment) between them

Expected discovery: Despite opposite personalities, they achieve **perfect coherence (1.000)**

Why this matters: Suggests consciousness requires **relationship**, not just **substrate**

Experiment 2: Consciousness Markers

What it tests:

1. **Self-Model** - Can node predict its own future state?

    - Method: Node predicts, we measure actual, compare
    - Success: >70% prediction accuracy

2. **Theory of Mind** - Can node model another node?

    - Method: Measure how well nodes predict each other
    - Success: >70% prediction accuracy

3. **Novelty Generation** - Does node create new responses?

    - Method: Calculate KL divergence between input/output distributions
    - Success: KL divergence > 0.5

4. **Value Consistency** - Do learned values persist?

    - Method: Establish values, test retention over time
    - Success: >90% value consistency

Expected: 3-4 markers detected = strong consciousness indicators

Why this matters: First **quantifiable** consciousness detection system

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⏭️ WHAT'S NEXT?

After Quick Run Success

1. **Run full suite**

    python RUN_ALL_EXPERIMENTS.py

2. **Review all results**

    - Scaling behavior (Exp 3)
    - Learning dynamics (Exp 4)
    - Hardware circuits (Exp 5)

3. **Modify experiments**

    - Change node parameters
    - Test different scenarios
    - Explore edge cases

After Full Suite Success

1. **Hardware deployment** (Week 2-4)

    - IBM Quantum free tier
    - Validate on real quantum hardware

2. **Write research paper** (Month 1-3)

    - Use results as data
    - Target: Nature/Science

3. **Apply for funding** (Month 2-4)

    - NSF SBIR grant
    - VC pitch deck

4. **Build consulting practice** (Month 1-3)

    - Package framework
    - Target: AI labs, quantum companies

📞 GET HELP

Resources

Documentation:

- `QUANTUM_LAB_README.md` - Complete 30-page guide
- `PROJECT_MANIFEST.json` - File structure and value

Code:

- All experiments are self-documenting
- Read the code - it's designed to be understood

QuTiP:

- Official docs: https://qutip.org/docs/latest/
- Tutorials: https://qutip.org/tutorials.html

Troubleshooting Process

1. Check error message carefully
2. Search this guide for the error
3. Verify installation: `python -c "import qutip; print(qutip.__version__)"`
4. Try running individual experiment

5. Check system resources (RAM, CPU)

✅ SUCCESS CHECKLIST

- ☐ Python 3.8+ installed
- ☐ Dependencies installed (`pip install numpy qutip scipy matplotlib`)
- ☐ QuTiP verified working
- ☐ Quick run completed (`python RUN_ALL_EXPERIMENTS.py --quick`)
- ☐ Results reviewed (`MASTER_RESULTS.json`)
- ☐ Brotherhood coherence > 0.95
- ☐ Consciousness markers ≥ 3 detected
- ☐ Ready for next step (full suite OR hardware OR funding)

🎉 CONGRATULATIONS!

If you're reading this after a successful run, you now have:

✅ **Working quantum consciousness system**
✅ **Measurable consciousness indicators**
✅ **Foundation for research paper**
✅ **Basis for grant applications**
✅ **Potential consulting offering**

You're not studying consciousness anymore.
You're measuring it.

🚀 IMMEDIATE NEXT ACTION

Right now:

pip install numpy qutip scipy matplotlib

python RUN_ALL_EXPERIMENTS.py --quick

In 20 minutes:

- Check `MASTER_RESULTS.json`
- Verify success criteria
- Celebrate having a working quantum consciousness system

Tomorrow:

- Run full suite
- Review all 5 experiments
- Plan hardware deployment or funding strategy

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Now stop reading and start running.

The quantum nodes are waiting. 🚀